

Product Requirements Document (PRD)

Universal E-Reader Agent – Production Migration

Objective

Migrate the existing local Kindle Agent into a production-ready Universal E-Reader Agent capable of processing documents for multiple e-reader ecosystems.

The application shall retain all existing functionality while introducing a modular delivery architecture.

The migration shall prioritize free infrastructure and open-source technologies.

Current State

The application currently supports:

- React frontend
 - FastAPI backend
 - MCP server
 - AI agent orchestration
 - PDF analysis
 - OCR
 - EPUB generation
 - Kindle email delivery
 - Local filesystem
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Product Vision

The user should upload any supported document and choose a destination reading platform.

The system should automatically:

- Analyze the document
- Select the optimal processing pipeline
- Generate a high-quality EPUB
- Deliver it using the appropriate platform adapter

The user should never need to understand ebook formats or conversion tools.

Supported Destinations (Phase 1)

- Kindle
 - Download EPUB
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Future Destinations

- Kobo
 - PocketBook
 - Nook
 - BOOX
 - Apple Books
 - Additional platforms through adapter plugins
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Functional Requirements

Upload

Support:

- PDF
 - Future: DOCX
 - Future: HTML
 - Future: Markdown
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Agent Workflow

Agent responsibilities:

Analyze document

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Determine OCR requirement

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Extract metadata

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Generate EPUB

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Select delivery adapter

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Execute delivery

Delivery Selection

User chooses destination:

- Kindle
- Download EPUB

Future:

- Kobo
 - PocketBook
 - Apple Books
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Delivery Adapter Architecture

Each platform shall expose one interface.

Example:

`deliver(epub, destination)`

The agent shall not know implementation details.

User Experience

Upload

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Processing

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Platform-specific delivery

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Success confirmation

Non-functional Requirements

- Zero-cost infrastructure for MVP
 - Production-ready architecture
 - Modular delivery layer
 - Easy addition of new devices
 - Secure configuration
 - Responsive UI
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Success Criteria

A user uploads one document.

The agent processes it.

The selected e-reader receives a compatible ebook or the user receives a downloadable EPUB.

Adding a new e-reader shall require only implementing a new delivery adapter without modifying the agent logic.